

Opening Range Breakout with Relative Volume Confirmation (ORB-RVOL)

Strategy Research Document (SRD)

Version

1.0

Strategy Type

Intraday Momentum

Market

NSE Equities (NIFTY 500 Universe Preferred)

Timeframe

5-Minute Bars

Holding Period

Intraday Only (No Overnight Positions)

1. Executive Summary

The Opening Range Breakout with Relative Volume Confirmation (ORB-RVOL) strategy is an intraday momentum strategy designed to capture directional institutional order flow generated by overnight information events and concentrated execution activity during the market open.

The strategy identifies stocks experiencing unusually high opening participation through Relative Volume (RVOL) analysis and enters positions only after a confirmed breakout from the opening range. By combining volume confirmation, volatility regime filtering, disciplined risk management, and strict intraday exits, the strategy seeks to capture strong directional moves while minimizing exposure to false breakouts.

2. Core Market Inefficiency

The strategy exploits the tendency of financial markets to underreact to overnight information.

Examples include:

- Earnings releases
- Corporate announcements
- Regulatory developments
- Global market movements
- Macroeconomic news
- Sector-specific events

When significant information arrives outside market hours, institutional participants frequently require substantial time to execute large orders after the market opens.

This creates:

- Elevated opening volume
- Persistent directional flows
- Delayed price discovery
- Momentum continuation throughout the trading session

The strategy identifies these situations through abnormal opening volume and participates only after directional confirmation.

3. Trading Universe

Eligible Securities

Preferred Universe:

- NIFTY 500 Stocks

Alternative Universe:

- NSE Large Cap Stocks
- NSE Mid Cap Stocks

Liquidity Requirements

Minimum Average Daily Value Traded (ADTV):

₹10 Crore

Minimum Average Daily Volume:

500,000 shares

Exclusions

Avoid:

- Illiquid securities
 - Stocks under ASM/GSM restrictions
 - Stocks with frequent circuit limits
 - Penny stocks
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4. Data Requirements

Real-Time Data

5-Minute OHLCV Data

Required Fields:

Field	Type
Timestamp	Datetime
Open	Float
High	Float
Low	Float

Close	Float
Volume	Integer

Historical Data

Minimum Lookback:

14 Trading Days

Preferred:

30 Trading Days

Required Historical Data:

- Daily OHLCV
 - 5-minute volume history
 - India VIX values
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5. Session Definitions

Event	Time (IST)
Market Open	09:15
Opening Range End	09:20
Entry Window Begins	09:20
Last Entry Time	14:30
Mandatory Exit	15:15

6. Opening Range Construction

The Opening Range (OR) is defined using the first five-minute candle of the trading session.

Time Interval:

09:15 → 09:20

Opening Range High

H_OR = High of first 5-minute candle

Opening Range Low

L_OR = Low of first 5-minute candle

Opening Range Width

ORW = H_OR - L_OR

Opening Range Percentage

ORW% = ((H_OR - L_OR) / Previous Day Close) × 100

7. Relative Volume (RVOL)

Objective

Measure whether current opening activity is significantly greater than normal participation.

Opening Volume

V_{today} = Volume of today's first 5-minute candle

Historical Baseline

Average volume of the first five-minute candle over the previous 14 trading sessions.

$\text{AvgVol14} = \text{Mean}(V1, V2, \dots V14)$

RVOL Calculation

$\text{RVOL} = V_{\text{today}} / \text{AvgVol14}$

Qualification Rule

$\text{RVOL} \geq 3.0$

Only stocks meeting this threshold become eligible for trading.

8. Volatility Regime Filter

The strategy performs poorly during extremely low-volatility environments where mean reversion dominates.

Input

India VIX Daily Value

Rule

Calculate the historical volatility percentile.

Trading is disabled whenever:

India VIX < Historical 20th Percentile

Example

If:

20th Percentile VIX = 11

and

Current VIX = 9.8

Then:

No trades are permitted.

9. Long Entry Conditions

A long trade is initiated only when all conditions are satisfied.

Condition 1

$RVOL \geq 3.0$

Condition 2

A 5-minute candle closes above the Opening Range High.

$Close > H_OR$

Condition 3

Entire candle remains above Opening Range.

$\text{Low} > \text{H_OR}$

Entry Execution

Enter at the open of the next candle.

Entry Price = Next Candle Open

10. Short Entry Conditions

A short trade is initiated only when all conditions are satisfied.

Condition 1

$\text{RVOL} \geq 3.0$

Condition 2

$\text{Close} < \text{L_OR}$

Condition 3

$\text{High} < \text{L_OR}$

Entry Execution

Enter at the open of the next candle.

Entry Price = Next Candle Open

11. Initial Stop Loss

The strategy uses ATR-based dynamic risk management.

ATR Calculation

Daily ATR(14)

True Range:

$$\text{TR} = \text{Max}(\text{High} - \text{Low}, |\text{High} - \text{Previous Close}|, |\text{Low} - \text{Previous Close}|)$$
$$\text{ATR14} = \text{Average}(\text{TR over 14 days})$$

Long Stop

$$\text{Stop} = \text{Entry} - (1.5 \times \text{ATR14})$$

Short Stop

$$\text{Stop} = \text{Entry} + (1.5 \times \text{ATR14})$$

12. Narrow Opening Range Exception

If:

$ORW\% < 0.75\%$

Use Opening Range boundaries instead of ATR stops.

Long Trade

Stop = L_OR

Short Trade

Stop = H_OR

13. Risk Unit (R)

Define:

$R = |\text{Entry} - \text{Stop}|$

This represents the initial trade risk.

Example

Entry = ₹500

Stop = ₹490

R = ₹10

14. Profit Management

Initial Objective

Target = $3R$

Long:

Target = Entry + $(3 \times R)$

Short:

Target = Entry - $(3 \times R)$

Dynamic Trailing System

At 1.5R Profit

Move stop to breakeven.

At 2R Profit

Trail stop using 1 ATR.

Beyond 2R

Continue ATR trailing until exit.

15. Failed Breakout Exit

Breakouts occasionally fail and re-enter the opening range.

To identify failed breakouts:

If the stock closes back inside the opening range for three consecutive bars:

Exit immediately.

Long Trade Failure

$L_OR < Close < H_OR$

for 3 consecutive bars.

Short Trade Failure

$L_OR < Close < H_OR$

for 3 consecutive bars.

16. End-of-Day Exit

All positions must be liquidated before market close.

Mandatory Exit Time:

15:15 IST

No overnight holding is permitted.

17. Position Sizing

Risk Per Trade

Maximum Risk:

1% of Total Capital

Example

Capital = ₹10,00,000

Maximum Risk = ₹10,000

Quantity Formula

Quantity = Risk Capital / R

Example

Risk Capital = ₹10,000

R = ₹5

Quantity = 2,000 Shares

18. Portfolio-Level Constraints

Maximum Open Positions

4 simultaneous positions

Sector Exposure Control

Maximum:

2 positions per sector

Examples:

- Banking
- IT
- Pharma
- Metals
- Auto

This reduces concentration risk.

19. Transaction Cost Assumptions

Backtesting must include:

Slippage

Entry Slippage: 0.05%

Exit Slippage: 0.05%

Round Trip Slippage: 0.10%

Trading Costs

Include:

- Brokerage
- STT
- Exchange Charges
- GST
- SEBI Charges
- Stamp Duty

Use realistic NSE intraday assumptions.

20. Performance Metrics

The backtesting engine must generate:

Return Metrics

- CAGR
 - Annualized Return
 - Monthly Return
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Risk Metrics

- Maximum Drawdown
 - Volatility
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Risk-Adjusted Metrics

- Sharpe Ratio
 - Sortino Ratio
 - Calmar Ratio
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Trade Statistics

- Win Rate
 - Profit Factor
 - Average Win
 - Average Loss
 - Expectancy
 - Payoff Ratio
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Execution Statistics

- Average Holding Time
 - Average Slippage
 - Trades Per Day
 - Capacity Utilization
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21. Recommended Enhancements

The following filters can materially improve robustness:

VWAP Confirmation

Longs only above VWAP.

Shorts only below VWAP.

Gap Filter

Trade only stocks with:

Absolute Gap $\geq 1\%$

Stronger RVOL Threshold

Increase threshold:

RVOL ≥ 4

Market Trend Filter

Longs only when NIFTY 50 > 20 EMA.

Shorts only when NIFTY 50 < 20 EMA.

Candidate Ranking

Rank eligible stocks by:

Score = RVOL $\times$ Gap%

Trade only the highest-ranked opportunities each session.

Expected Strategy Characteristics

Holding Period:

30 Minutes – 5 Hours

Average Trades:

1–4 Trades Per Day

Expected Win Rate:

40–55%

Target Profit Factor:

1.5–2.0+

Target Sharpe Ratio:

1.2–2.0+

Primary Risk:

False breakouts during low-volatility or mean-reverting environments.

Primary Edge:

Institutional participation identified through abnormal opening volume and directional opening-range expansion.